Signals and Systems
Assignment-3: Solution

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Exercise 1 (1)

- Check the orthogonality by evaluating the following integral:

$$
\begin{aligned}
I &= \int_{-1/2}^{1/2} f_1(t) f_2^*(t) \mathrm{d}t = AB \int_{-1/2}^{1/2} \sin(4\pi t)\sin(2\pi t)\mathrm{d}t \\
&= \frac{AB}{2}\int_{-1/2}^{1/2} \cos(2\pi t)\mathrm{d}t - \frac{AB}{2}\int_{-1/2}^{1/2} \cos(6\pi t)\mathrm{d}t \\
&= \frac{AB}{2}\frac{1}{2\pi}\left[\underbrace{\sin\left(2\pi\frac{1}{2}\right)}_{=0} - \underbrace{\sin\left(-2\pi\frac{1}{2}\right)}_{=0}\right] \\
&\quad - \frac{AB}{2}\frac{1}{6\pi}\left[\underbrace{\sin\left(6\pi\frac{1}{2}\right)}_{=0} - \underbrace{\sin\left(-6\pi\frac{1}{2}\right)}_{=0}\right] \\
&= 0
\end{aligned}
$$

Exercise 2 (1)

- The function $g(t)$ can be written as:

$$
\begin{aligned}
g(t) &= (1+j)e^{j4\pi t} + (1-j)e^{-j4\pi t} \\
&= e^{j4\pi t} + e^{-j4\pi t} + j(e^{j4\pi t} - e^{-j4\pi t}) \\
&= 2\frac{e^{j4\pi t} + e^{-j4\pi t}}{2} - 2\frac{e^{j4\pi t} - e^{-j4\pi t}}{2j} \\
&= 2\cos(4\pi t) - 2\sin(4\pi t)
\end{aligned}
$$

Exercise 3.1 (1)

- **First Method**:

$$
\begin{aligned}
x(t) &= 2\cos(100\pi(t-0.005)) \\
&= e^{j100\pi(t-0.005)} + e^{-j100\pi(t-0.005)} \\
&= \underbrace{e^{-0.5\pi j}}_{=X[1]} e^{j100\pi t} + \underbrace{e^{0.5\pi j}}_{=X[-1]} e^{-j100\pi t}
\end{aligned}
$$

  - $X[1] = \cos(0.5\pi) - j\sin(0.5\pi) = -j$.
  - $X[-1] = \cos(0.5\pi) + j\sin(0.5\pi) = j$ (note that $X[-1] = X^*[1]$).
  - $X[k] = 0$ for $k \neq \pm 1$.

Exercise 3.1 (2)

- **Second Method**:

$$x(t) = 2\cos(100\pi(t - 0.005)) = 2\cos(100\pi t - 0.5\pi)$$
$$= \underbrace{2}_{=X_s[1]} \sin(100\pi t)$$

  - $X[1] = \frac{X_c[1] - jX_s[1]}{2} = -j$.
  - $X[-1] = X^*[1] = j$.
  - $X[k] = 0$ for $k \neq \pm 1$.

Exercise 3.1 (3)

- In the time domain:

$$P_{av} = \frac{2^2}{2} = 2 \quad \mathrm{W}$$

since the average power of the sinusoid $A\cos(2\pi ft + \theta)$ is $A^2/2$.

- In the frequency domain:

$$\begin{aligned} P_{av} &= \sum_{k=-\infty}^{+\infty} |X[k]|^2 \\ &= |X[-1]|^2 + |X[1]|^2 = |j|^2 + |-j|^2 = 1 + 1 = 2 \quad \mathrm{W} \end{aligned}$$

resulting in the same answer.

Exercise 3.2 (1)

- **First Method:**

$$
\begin{aligned}
x(t) &= \sin(50\pi t - \pi/3) \\
&= \frac{e^{j(50\pi t - \pi/3)} - e^{-j(50\pi t - \pi/3)}}{2j} \\
&= \underbrace{\frac{1}{2j} e^{-j\pi/3}}_{=X[1]} e^{j50\pi t} \underbrace{- \frac{1}{2j} e^{j\pi/3}}_{=X[-1]} e^{-j50\pi t}
\end{aligned}
$$

where:

$$X[1] = -\frac{j}{2}\left[\cos(\pi/3) - j\sin(\pi/3)\right] = -\frac{j}{2}\left[\frac{1}{2} - j\frac{\sqrt{3}}{2}\right] = -\frac{\sqrt{3}}{4} - \frac{1}{4}j$$

$$X[-1] = \frac{j}{2}\left[\cos(\pi/3) + j\sin(\pi/3)\right] = \frac{j}{2}\left[\frac{1}{2} + j\frac{\sqrt{3}}{2}\right] = -\frac{\sqrt{3}}{4} + \frac{1}{4}j$$

$$X[k] = 0 \quad ; \quad k \neq \pm 1$$

- Note that $X[-1] = X^*[1]$.

Exercise 3.2 (2)

- **Second Method:**

$$
\begin{aligned}
x(t) &= \sin(50\pi t - \pi/3) \\
&= \underbrace{\cos(\pi/3)}_{=X_s[1]=1/2} \sin(50\pi t) \underbrace{-\sin(\pi/3)}_{=X_c[1]=-\sqrt{3}/2} \cos(50\pi t)
\end{aligned}
$$

Consequently:

$$
X[1] = \frac{X_c[1] - jX_s[1]}{2} = \frac{1}{2}\left[-\frac{\sqrt{3}}{2} - j\frac{1}{2}\right] = -\frac{\sqrt{3}}{4} - \frac{1}{4}j
$$

$$
X[-1] = X^*[1] = -\frac{\sqrt{3}}{4} + \frac{1}{4}j
$$

$$
X[k] = 0 \quad ; \quad k \neq \pm 1
$$

- Note that $x(t)$ is neither even nor odd implying that the Fourier series coefficients are neither real nor pure imaginary.

Exercise 3.2 (3)

- In the time domain:

$$P_{av} = \frac{1^2}{2} = 0.5 \quad \mathrm{W}$$

since the average power of the sinusoid $A\sin(2\pi ft + \theta)$ is $A^2/2$.

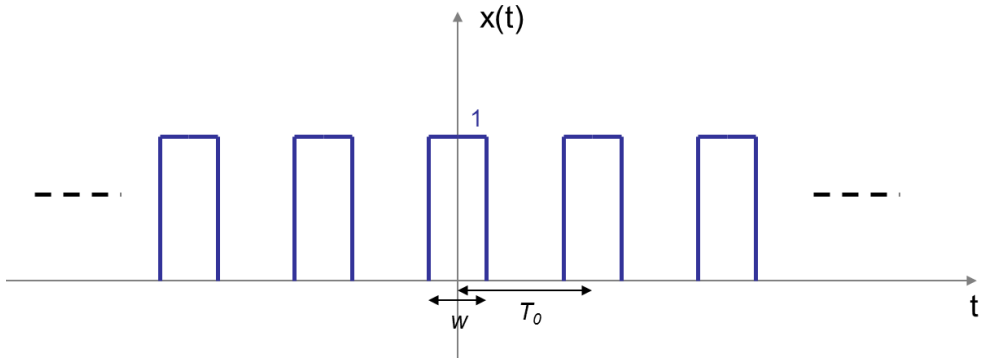
- In the frequency domain:

$$\begin{aligned} P_{av} &= \sum_{k=-\infty}^{+\infty} |X[k]|^2 = |X[-1]|^2 + |X[1]|^2 \\ &= \left| -\frac{1}{2j} e^{j\pi/3} \right|^2 + \left| \frac{1}{2j} e^{-j\pi/3} \right|^2 = \frac{1}{4} + \frac{1}{4} = 0.5 \quad \mathrm{W} \end{aligned}$$

resulting in the same answer (Note that $|e^{j\theta}| = 1$).

Exercise 4 (1)

- Consider the CT signal shown in the figure below.



- In lecture 8, it was proven that the Fourier Series (FS) coefficients of this signal are given by:

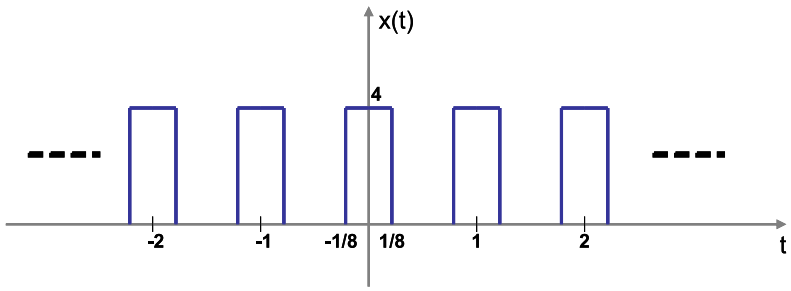
$$X[k] = \frac{w}{T_0}\text{sinc}\left(\frac{w}{T_0}k\right) \quad ; \quad k \in \mathbb{Z}$$

  - This signal will be referred to as the reference signal in what follows.
- From lecture 8, the periodic repetition of the signal $f(t)$ with period T_0 can be written as:

$$x(t) * \frac{1}{T_0}\text{comb}\left(\frac{t}{T_0}\right)$$

Exercise 4 (2)

- $x(t) = 4\mathrm{rect}(4t) * \mathrm{comb}(t)$ corresponds to the repetition of $4\mathrm{rect}(4t)$ with period $T_0 = 1$.
- $x(t)$ is depicted in the figure below.



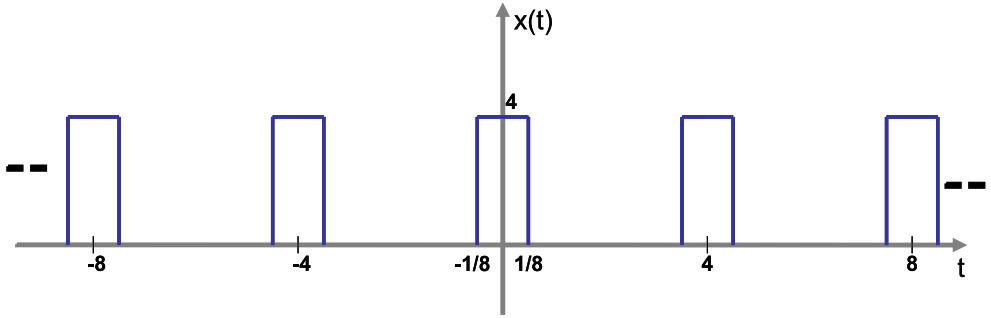
- We observe that $x(t)$ corresponds to the reference signal (multiplied by 4) with $T_0 = 1$ and $w = 1/4$. Consequently:

$$X[k] = 4\frac{w}{T_0}\mathrm{sinc}\left(\frac{w}{T_0}k\right) = \mathrm{sinc}\left(\frac{k}{4}\right)$$

  - Note that $X[k] = 0$ when $k/4$ is a nonzero integer implying that k is a nonzero multiple of 4 ($k = \pm 4, \pm 8, \pm 12, \ldots$).

Exercise 4 (3)

- $x(t) = \text{rect}(4t) * \text{comb}\left(\frac{t}{4}\right) = 4\text{rect}(4t) * \frac{1}{4}\text{comb}\left(\frac{t}{4}\right)$ corresponds to the repetition of $4\text{rect}(4t)$ with period $T_0 = 4$.
- $x(t)$ is depicted in the figure below.



- We observe that $x(t)$ corresponds to the reference signal (multiplied by 4) with $T_0 = 4$ and $w = 1/4$. Consequently:

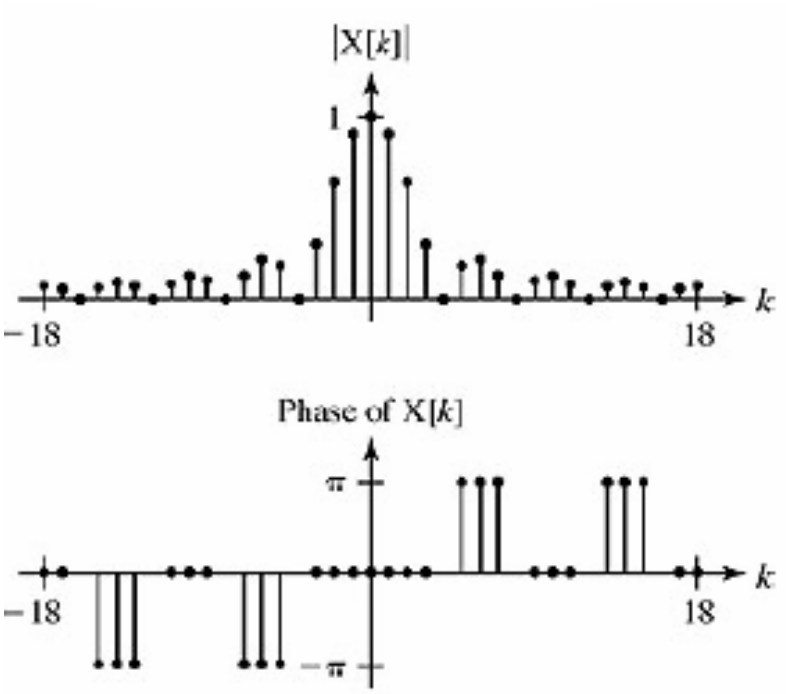
$$X[k] = 4\frac{w}{T_0}\text{sinc}\left(\frac{w}{T_0}k\right) = \frac{1}{4}\text{sinc}\left(\frac{k}{16}\right)$$

  - Note that $X[0] = 1/4$.
  - Note that $X[k] = 0$ when k is a nonzero multiple of 16.

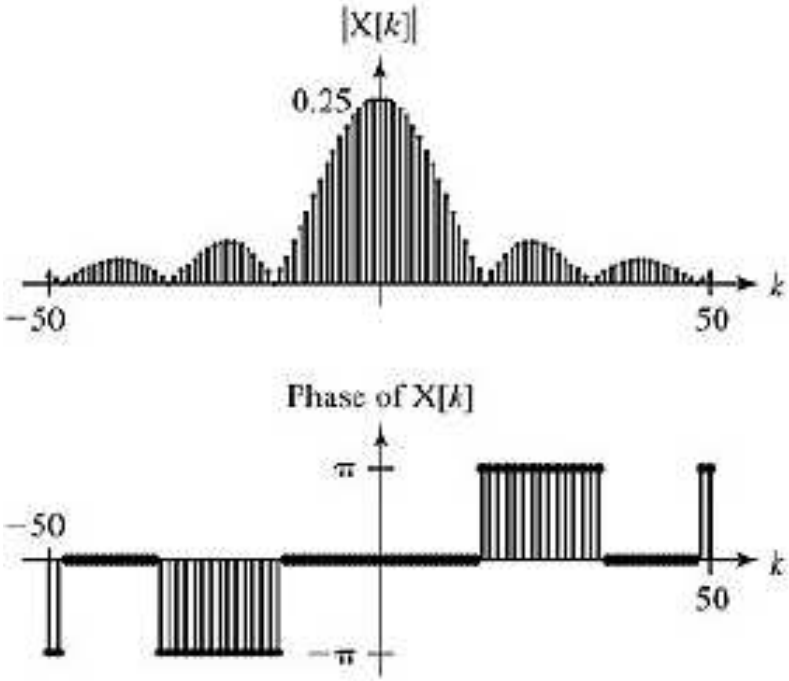
Exercise 4 (4)

- For the above signals, the magnitudes and phases of $X[k]$ are plotted in the figures below versus k.

$x(t) = 4\text{rect}(4t) * \text{comb}(t)$



$x(t) = 4\text{rect}(4t) * \frac{1}{4}\text{comb}\left(\frac{t}{4}\right)$



Exercise 5 (1)

- Since $x(t)$ is odd:

$$X_c[k] = 0 \;\; ; \;\; k = 0, 1, 2, \ldots$$

- On the other hand:

$$\begin{aligned} X_s[k] &= \frac{2}{T_0} \int_{-T_0/2}^{T_0/2} x(t) \sin(2\pi(kf_0)t) \mathrm{d}t \\ &= \frac{4}{T_0} \int_{0}^{T_0/2} x(t) \sin(2\pi(kf_0)t) \mathrm{d}t \end{aligned}$$

  - where the second equality follows since $x(t)$ is odd and $\sin(2\pi(kf_0)t)$ is odd implying that $x(t)\sin(2\pi(kf_0)t)$ is even.

- Consequently:

$$\begin{aligned} X_s[k] &= \frac{4}{T_0} \int_{0}^{T_0/2} A \sin(2\pi(kf_0)t) \mathrm{d}t \\ &= \frac{4A}{T_0} \frac{1}{2\pi k f_0} \left[-\cos(2\pi(kf_0)t) \right]_0^{T_0/2} \\ &= \frac{2A}{\pi k T_0 f_0} \left[1 - \cos\left(2\pi(kf_0) \frac{T_0}{2} \right) \right] \end{aligned}$$

Exercise 5 (2)

- Given that $f_0 = 1/T_0$ implying that $f_0 T_0 = 1$:

$$X_s[k] = \frac{2A}{\pi k}\left[1 - \cos(\pi k)\right] = \frac{2A}{\pi k}\left[1 - (-1)^k\right]$$

- Consequently, for $k > 0$:

$$\begin{aligned} X[k] &= \frac{X_c[k] - jX_s[k]}{2} = -j\frac{X_s[k]}{2} \\ &= -j\frac{A}{\pi k}\left[1 - (-1)^k\right] = \begin{cases} -j\frac{2A}{k\pi}, & k \text{ is odd;} \\ 0, & k \text{ is even.} \end{cases} \end{aligned}$$

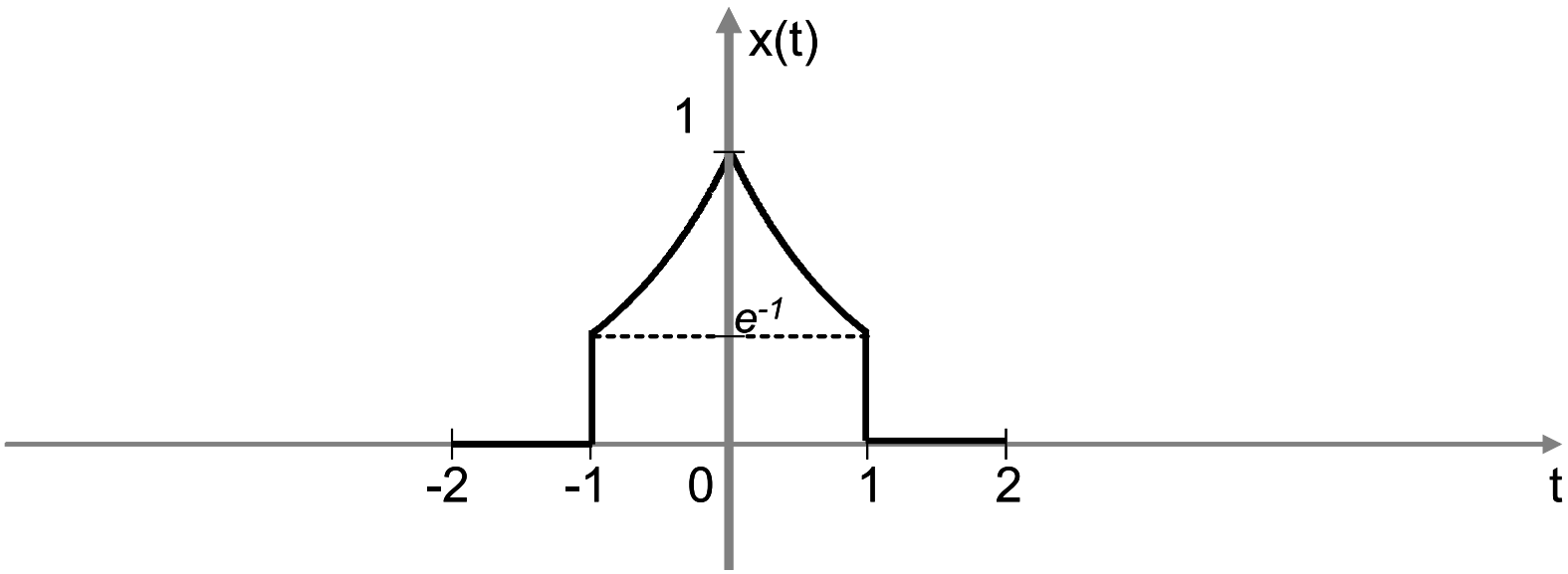
- Finally, since $X[-k] = X^*[k]$, then for $k < 0$:

$$X[k] = X^*[-k] = \begin{cases} -j\frac{2A}{k\pi}, & k \text{ is odd;} \\ 0, & k \text{ is even.} \end{cases}$$

Exercise 6 (1)

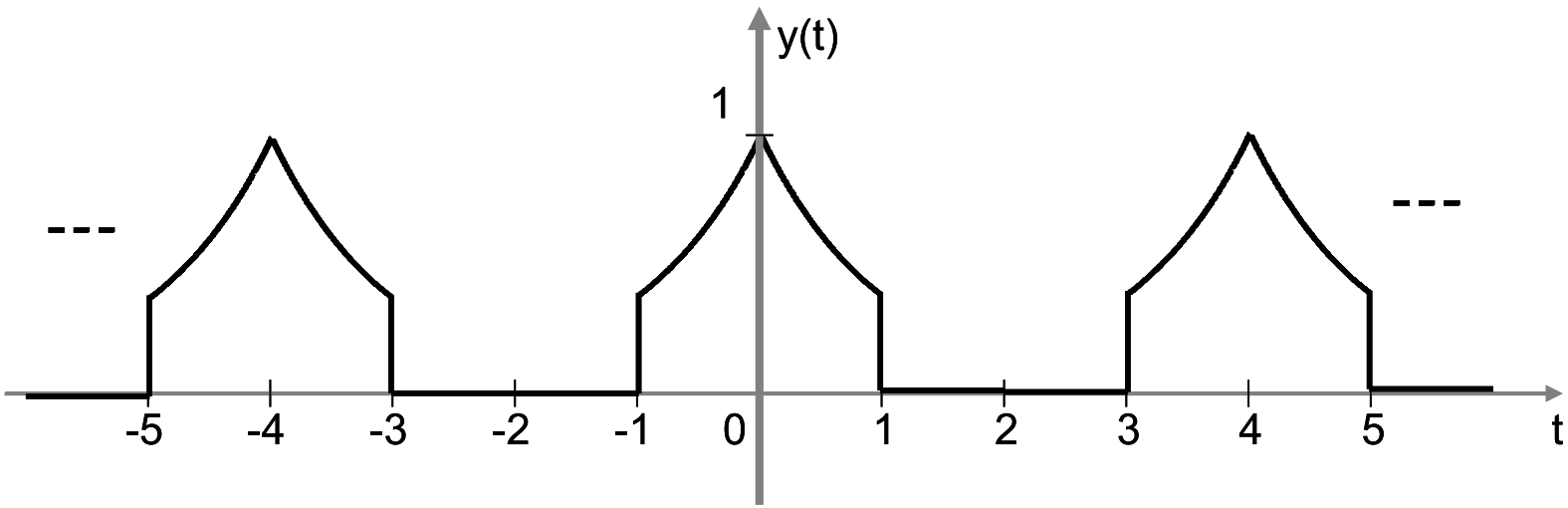
$x(t)$ is included in the time interval $[-2 \;\; 2]$.

$$x(t) = \begin{cases} 0, & -2 \leq t \leq -1; \\ e^{-|t|} = e^{t}, & -1 \leq t \leq 0; \\ e^{-|t|} = e^{-t}, & 0 \leq t \leq 1; \\ 0, & 1 \leq t \leq 2. \end{cases}$$



Exercise 6 (2)

$y(t)$ is the periodic extension of $x(t)$:



- It can be written as:

$$y(t) = x(t) * \frac{1}{T_0}\text{comb}\left(\frac{t}{T_0}\right) \quad ; \quad T_0 = 4 \text{ is the period}$$
$$= x(t) * \frac{1}{4}\text{comb}\left(\frac{t}{4}\right)$$

Exercise 6 (3)

The k-th coefficient of the Fourier series is given by:

$$Y[k] = \frac{1}{T_0} \int_{T_0} y(t) e^{-j2\pi(kf_0)t} \mathrm{d}t$$

where $T_0 = 4$ and $f_0 = \frac{1}{T_0} = \frac{1}{4}$. Therefore:

$$\begin{aligned}
Y[k] &= \frac{1}{4} \int_{-2}^{+2} y(t) e^{-jk\frac{\pi}{2}t} \mathrm{d}t = \frac{1}{4} \int_{-1}^{+1} y(t) e^{-jk\frac{\pi}{2}t} \mathrm{d}t \\
&= \frac{1}{4} \left[\int_{-1}^{0} e^{t} e^{-jk\frac{\pi}{2}t} \mathrm{d}t + \int_{0}^{+1} e^{-t} e^{-jk\frac{\pi}{2}t} \mathrm{d}t \right] \\
&= \frac{1}{4} \left[\frac{1}{1 - jk\frac{\pi}{2}} \left[e^{t} e^{-jk\frac{\pi}{2}t} \right]_{-1}^{0} - \frac{1}{1 + jk\frac{\pi}{2}} \left[e^{-t} e^{-jk\frac{\pi}{2}t} \right]_{0}^{1} \right] \\
&= \frac{1}{4} \left[\frac{1}{1 - jk\frac{\pi}{2}} \left[1 - e^{-1} e^{jk\frac{\pi}{2}} \right] - \frac{1}{1 + jk\frac{\pi}{2}} \left[e^{-1} e^{-jk\frac{\pi}{2}} - 1 \right] \right]
\end{aligned}$$

Exercise 6 (4)

$$
\begin{aligned}
Y[k] &= \frac{1}{4}\left[\frac{1}{1-jk\frac{\pi}{2}} + \frac{1}{1+jk\frac{\pi}{2}} - e^{-1}\left[\frac{e^{jk\frac{\pi}{2}}}{1-jk\frac{\pi}{2}} + \frac{e^{-jk\frac{\pi}{2}}}{1+jk\frac{\pi}{2}}\right]\right] \\
&= \frac{1}{4}\left[\frac{2}{1+\left(k\frac{\pi}{2}\right)^2} - \frac{e^{-1}}{1+\left(k\frac{\pi}{2}\right)^2}\left[e^{jk\frac{\pi}{2}} + e^{-jk\frac{\pi}{2}} + jk\frac{\pi}{2}\left[e^{jk\frac{\pi}{2}} - e^{-jk\frac{\pi}{2}}\right]\right]\right] \\
&= \frac{1}{4+k^2\pi^2}\left[2 - e^{-1}\left[2\cos\left(k\frac{\pi}{2}\right) - \pi k\sin\left(k\frac{\pi}{2}\right)\right]\right]
\end{aligned}
$$

- When k is odd, $\cos\left(k\frac{\pi}{2}\right) = 0$ and:

$$
Y[k] = \frac{2}{4+k^2\pi^2}\left[1 + e^{-1}k\frac{\pi}{2}\sin\left(k\frac{\pi}{2}\right)\right]
$$

- When k is even, $\sin\left(k\frac{\pi}{2}\right) = 0$ and:

$$
Y[k] = \frac{2}{4+k^2\pi^2}\left[1 - e^{-1}\cos\left(k\frac{\pi}{2}\right)\right]
$$

Exercise 6 (5)

Determine the trigonometric Fourier series representation:

- $Y_s[k]$ can be obtained from $Y[k]$ by applying:

$$Y_s[k] = -2\mathrm{Im}\{Y[k]\} = 0 \;\; ; \;\; k = 1, 2, \ldots$$

- $Y_c[k]$ can be obtained from $Y[k]$ by applying:

$$Y_c[k] = 2\mathrm{Re}\{Y[k]\} = 2Y[k] \;\; ; \;\; k = 0, 1, 2, \ldots$$

As a conclusion:

$$Y_c[k] = \begin{cases} \frac{4}{4+k^2\pi^2}\left[1 + e^{-1}k\frac{\pi}{2}\sin\left(k\frac{\pi}{2}\right)\right], & k \text{ is odd;} \\ \frac{4}{4+k^2\pi^2}\left[1 - e^{-1}\cos\left(k\frac{\pi}{2}\right)\right], & k \text{ is even.} \end{cases}$$

Note that the function is even. Consequently, $Y[k]$ is real and $Y_s[k] = 0$ for all values of k.

Exercise 6 (6)

What is the power of the third harmonic?

- The power of the third harmonic is given by:

$$
\begin{aligned}
P_3 &= 2|Y[3]|^2 \\
&= 2\frac{2^2}{(4+k^2\pi^2)^2}\left[1+e^{-1}k\frac{\pi}{2}\sin\left(k\frac{\pi}{2}\right)\right]^2 \quad ; \quad k=3 \\
&= \frac{8}{(4+9\pi^2)^2}\left[1-e^{-1}3\frac{\pi}{2}\right]^2 \\
&= \cdots \\
&= 4.9964\ 10^{-4}\ \ \mathrm{W}
\end{aligned}
$$

Exercise 7.1

- For $k \neq 0$, $X[-k]$ takes the following value:

$$X[-k] = 2\frac{\sin\left(-k\frac{\pi}{2}\right)}{j(-k\pi)^2}e^{-j(-k)\frac{\pi}{2}} = -2\frac{\sin\left(k\frac{\pi}{2}\right)}{j(k\pi)^2}e^{jk\frac{\pi}{2}}$$

- On the other hand, $X[k]^*$ takes the following value:

$$X[k]^* = \left[2\frac{\sin\left(k\frac{\pi}{2}\right)}{j(k\pi)^2}e^{-jk\frac{\pi}{2}}\right]^* = 2\frac{\sin\left(k\frac{\pi}{2}\right)}{-j(k\pi)^2}e^{jk\frac{\pi}{2}}$$

- Since $X[-k] = X[k]^*$ for all values of k, then the signal $x(t)$ is real (note that for $k = 0$, $X[0] = X[0]^* = \frac{1}{2}$).

Exercise 7.2

- For even values of k, $k\frac{\pi}{2}$ is an integer multiple of π implying that $\sin\left(k\frac{\pi}{2}\right) = 0$. Consequently:

$$X[k] = 0 \;\; ; \;\; k = \pm 2, \pm 4, \ldots$$

- For odd values of k, $e^{-jk\frac{\pi}{2}}$ is equal to either $-j$ or j. Consequently, $X[k]$ can be written as:

$$X[k] = 2\frac{\sin\left(k\frac{\pi}{2}\right)}{j(k\pi)^2}(\pm j) = \pm 2\frac{\sin\left(k\frac{\pi}{2}\right)}{(k\pi)^2} \;\; ; \;\; k = \pm 1, \pm 3, \ldots$$

- Consequently:

$$X[k] = \begin{cases} 1/2, & k = 0; \\ 0, & k = \pm 2, \pm 4, \ldots; \\ \pm 2\frac{\sin\left(k\frac{\pi}{2}\right)}{(k\pi)^2}, & k = \pm 1, \pm 3, \ldots. \end{cases}$$

implying that $X[k]$ is real. Consequently, $X_s[k] = 0$ for all values of k and $x(t)$ is even.

Exercise 7.3 (1)

- Denote by $f_0 = \frac{1}{T_0} = \frac{1}{4}$ the fundamental frequency of $x(t)$.
- Since the coefficient $X[k]$ corresponds to the frequency kf_0, the frequency components of $x(t)$ that will pass through the filter must satisfy the following relation:

$$-\frac{1}{3} \leq kf_0 = \frac{k}{4} \leq \frac{1}{3} \quad \Rightarrow \quad -\frac{4}{3} \leq k \leq \frac{4}{3}$$
$$\Rightarrow \quad k = -1, 0, 1$$

- Denote by $Y[k]$ the FS coefficients of the output:

$$Y[0] = X[0] = \frac{1}{2}$$
$$Y[1] = X[1] = 2\frac{\sin\left(\frac{\pi}{2}\right)}{j\pi^2}e^{-j\frac{\pi}{2}} = \frac{2(-j)}{j\pi^2} = -\frac{2}{\pi^2}$$
$$Y[-1] = X[-1] = 2\frac{\sin\left(-\frac{\pi}{2}\right)}{j(-\pi)^2}e^{j\frac{\pi}{2}} = \frac{2(-1)j}{j\pi^2} = -\frac{2}{\pi^2}$$
$$Y[k] = 0 \ ; \ \ k \neq 0, \pm 1$$

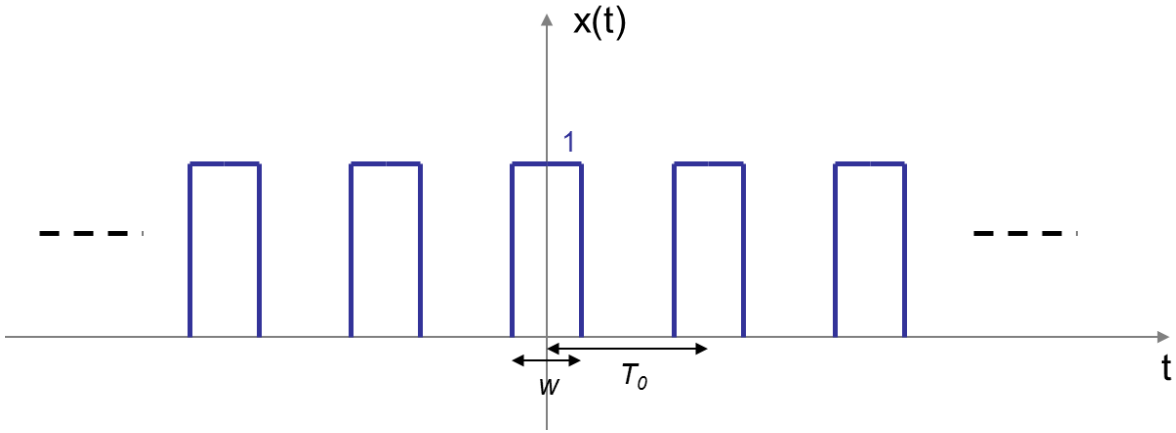
Exercise 7.3 (2)

- Consequently:

$$
\begin{aligned}
y(t) &= Y[0] + Y[1]e^{j2\pi\frac{1}{4}t} + Y[-1]e^{-j2\pi\frac{1}{4}t} \\
&= \frac{1}{2} - \frac{2}{\pi^2}\left[e^{j\frac{\pi t}{2}} + e^{-j\frac{\pi t}{2}}\right] \\
&= \frac{1}{2} - \frac{4}{\pi^2}\cos\left(\frac{\pi t}{2}\right)
\end{aligned}
$$

Exercise 8 (1)

- Consider the rectangular-pulse sequence shown in the figure below:



its FS coefficients are given by:

$$X[k] = \frac{w}{T_0}\text{sinc}\left(\frac{w}{T_0}k\right) \quad \forall \ k$$

- For $T_0 = aw$, $a \geq 1$ and:

$$X[k] = \frac{1}{a}\text{sinc}\left(\frac{k}{a}\right) \quad \forall \ k$$

Exercise 8 (2)

- Calculating the average power of $x(t)$ in the time domain:

$$
\begin{aligned}
P_{av} &= \frac{1}{T_0}\int_{T_0} x^2(t)\mathrm{d}t = \frac{1}{T_0}\int_{-T_0/2}^{T_0/2} x^2(t)\mathrm{d}t \\
&= \frac{2}{T_0}\int_{0}^{T_0/2} x^2(t)\mathrm{d}t \qquad (x(t) \text{ is even}) \\
&= \frac{2}{T_0}\int_{0}^{w/2} (1)^2\mathrm{d}t + \frac{2}{T_0}\int_{w/2}^{T_0/2} (0)^2\mathrm{d}t = \frac{w}{T_0} = \frac{1}{a}
\end{aligned}
$$

- Applying Parseval's theorem:

$$
P_{av} = \sum_{k=-\infty}^{+\infty} |X[k]|^2 = \frac{1}{T_0}\int_{T_0} x^2(t)\mathrm{d}t
$$

$$
\Rightarrow \sum_{k=-\infty}^{+\infty} \frac{1}{a^2}\mathrm{sinc}^2\left(\frac{k}{a}\right) = \frac{1}{a} \quad \Rightarrow \quad \sum_{k=-\infty}^{+\infty} \mathrm{sinc}^2\left(\frac{k}{a}\right) = a
$$

Exercise 9 (1)

- The fundamental frequency of $x(t)$ is:
  $f_0 = \gcd(25, 50, 75) = 25$ Hz.
- In terms of f_0, $x(t)$ can be written as:

$$x(t) = \underbrace{1}_{=X_c[1]} \cos(2\pi(f_0)t) + \underbrace{2}_{=X_s[2]} \sin(2\pi(2f_0)t) + \underbrace{1}_{=X_c[3]} \cos(2\pi(3f_0)t)$$

- Consequently, the exponential FS coefficients of $x(t)$ are:

$$X[1] = \frac{1}{2} \;\; ; \;\; X[2] = -j \;\; ; \;\; X[3] = \frac{1}{2}$$
$$X[-1] = \frac{1}{2} \;\; ; \;\; X[-2] = j \;\; ; \;\; X[-3] = \frac{1}{2}$$
$$X[k] = 0 \;\; ; \;\; k \neq \pm 1, \pm 2, \pm 3$$

Exercise 9 (2)

- The frequency components that will be <u>eliminated</u> by the filter must satisfy:

$$-f_c < kf_0 < f_c \;\Rightarrow\; -\frac{f_c}{f_c} < k < \frac{f_c}{f_c} \;\Rightarrow\; -\frac{60}{25} < k < \frac{60}{25}$$
$$\Rightarrow\; -2.4 < k < 2.4 \;\Rightarrow\; k \in \{0, \pm 1, \pm 2\}$$

- Consequently, the exponential FS coefficients of $y(t)$ are given by:

$$Y[3] = X[3] = \frac{1}{2}\;;\;\; Y[-3] = X[-3] = \frac{1}{2}\;;\;\; Y[k] = 0\,;\; k \neq \pm 3$$

- Therefore, $y(t)$ can be written as:

$$y(t) = \frac{1}{2}e^{j2\pi(3f_0)t} + \frac{1}{2}e^{-j2\pi(3f_0)t} = \cos(2\pi(3f_0)t) = \cos(150\pi t)$$

Exercise 9 (3)

- The power of $x(t)$ is:

$$P_x = |X[0]|^2 + 2\sum_{k=1}^{+\infty} |X[k]|^2$$
$$= 2\left[|X[1]|^2 + |X[2]|^2 + |X[3]|^2\right] = 2\left[\frac{1}{4} + 1 + \frac{1}{4}\right] = 3 \text{ (W)}$$

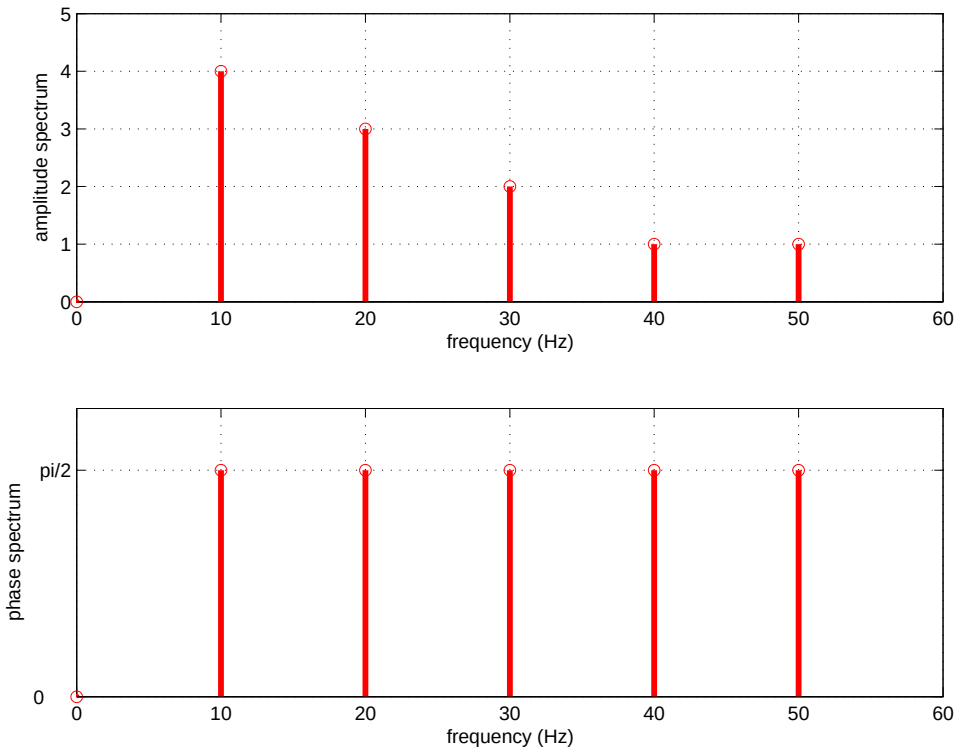
- The power of $y(t)$ is:

$$P_y = |Y[0]|^2 + 2\sum_{k=1}^{+\infty} |Y[k]|^2 = 2\left[|Y[3]|^2\right] = 2\left[\frac{1}{4}\right] = 0.5 \text{ (W)}$$

- The percentage of the input power appearing at the output of the filter is:

$$\frac{P_y}{P_x} \times 100\% = 16.67\%$$

Exercise 10 (1)



- From the spectrum of the signal, we can see that 2 consecutive harmonics are separated by 10 Hz. Consequently:

$$f_0 = 10 \text{ Hz} \quad \Rightarrow \quad T_0 = \frac{1}{f_0} = 0.1 \text{ s}$$

Exercise 10 (2)

- From the one-sided spectrum of the signal, we obtain the following values of the exponential FS coefficients:

$$
\begin{aligned}
X[0] &= |X[0]|e^{j\arg(X[0])} = 0e^0 = 0 \\
X[1] &= |X[1]|e^{j\arg(X[1])} = 4e^{j\frac{\pi}{2}} = 4j \\
X[2] &= |X[2]|e^{j\arg(X[2])} = 3e^{j\frac{\pi}{2}} = 3j \\
X[3] &= |X[3]|e^{j\arg(X[3])} = 2e^{j\frac{\pi}{2}} = 2j \\
X[4] &= |X[4]|e^{j\arg(X[4])} = 1e^{j\frac{\pi}{2}} = j \\
X[5] &= |X[5]|e^{j\arg(X[5])} = 1e^{j\frac{\pi}{2}} = j \\
X[k] &= 0 \quad k = 6, 7, 8, \ldots
\end{aligned}
$$

- From the property $X[-k] = X^*[k]$, we conclude:

$$
\begin{aligned}
&X[-1] = -4j \; ; \; X[-2] = -3j \\
&X[-3] = -2j \; ; \; X[-4] = -j \; ; \; X[-5] = -j \\
&X[-k] = 0 \quad k = 6, 7, 8, \ldots
\end{aligned}
$$

Exercise 10 (3)

- Determine wether $x(t)$ is even, odd or neither.

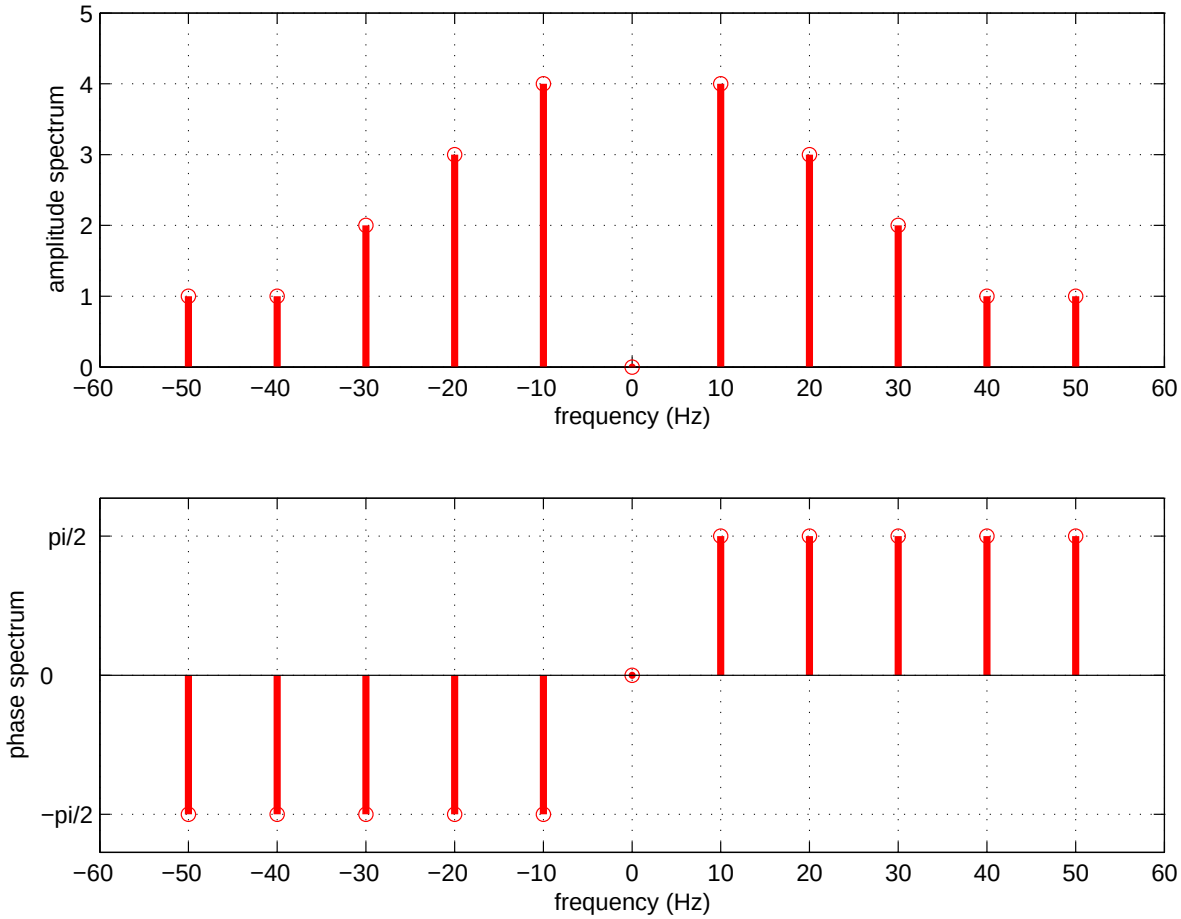
$$X[k] = \frac{X_c[k] - jX_s[k]}{2}$$

  - Since all the values $X[k]$ have pure imaginary parts, then $X_c[k] = 0$ for all values of k. Consequently, the signal is odd.

Exercise 10 (4)

- The spectrum can be obtained from the one-sided spectrum by applying the relation:

$$X[-k] = X^*[k] \quad \Rightarrow \quad \begin{cases} |X[-k]| = |X^*[k]| \\ \arg(X[-k]) = -\arg(X[k]) \end{cases}$$



Exercise 10 (5)

Determine the trigonometric Fourier series.

- Since we have proven that $x(t)$ is odd, then:

$$X_c[k] = 0 \quad k = 0, 1, 2, \ldots$$

- $X_s[k]$ can be obtained from:

$$X_s[k] = -2\mathrm{Im}\{X[k]\} \;\; k = 1, 2, \ldots$$

resulting in:

$$\begin{aligned} &X_s[1] = -8 \;\; ; \;\; X_s[2] = -6 \;\; ; \;\; X_s[3] = -4 \\ &X_s[4] = -2 \;\; ; \;\; X_s[5] = -2 \\ &X_s[k] = 0 \quad k = 6, 7, 8, \ldots \end{aligned}$$

Exercise 10 (6)

What is the average value of $x(t)$?

- The average value of $x(t)$ is given by:

$$\frac{1}{T_0}\int_{T_0} x(t)\mathrm{d}t = X[0] = 0$$

What is the power of $x(t)$?

- From Parseval' theorem:

$$\begin{aligned} P &= |X[0]|^2 + 2\sum_{k\in\mathbb{Z}^+} |X[k]|^2 \\ &= |X[0]|^2 + 2|X[1]|^2 + 2|X[2]|^2 + 2|X[3]|^2 + 2|X[4]|^2 + 2|X[5]|^2 \\ &= 0 + 2(16+9+4+1+1) = 62 \ \ \mathrm{W} \end{aligned}$$

Which harmonics carry the most power?

- From the magnitude spectrum, we can directly see that most of the power is contained in the first harmonic ($k = 1$).

Exercise 10 (7)

Determine the DC power:

$$P_0 = |X[0]|^2 = 0 \ \ \mathrm{W}$$

Determine the power of the third harmonic:

$$P_3 = 2|X[3]|^2 = 2 \times 4 = 8 \ \ \mathrm{W}$$